"I affirm that I have not given or received any unauthorized help on this report and that this work is my own."

Wideband amplifier with an automatic gain control

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221202584

Table 1 Output table

	V_{inp} (mV)	V_{outp} (V)	LED ON/OFF	I_{CC} (mA)
0.2MHz	10	0.9	ON	15
0.2MHz	V_{inpmin}= 10	1±0.1	ON	15
0.2MHz	100	0.9	ON	15
0.2MHz	V_{inpmax}= 500	1±0.1	ON	15
2MHz	10	0.92	ON	15
2MHz	V_{inpmin}= 10	1±0.1	ON	15
2MHz	100	0.92	ON	15
2MHz	V_{inpmax}= 6000	1± 0.1	ON	15

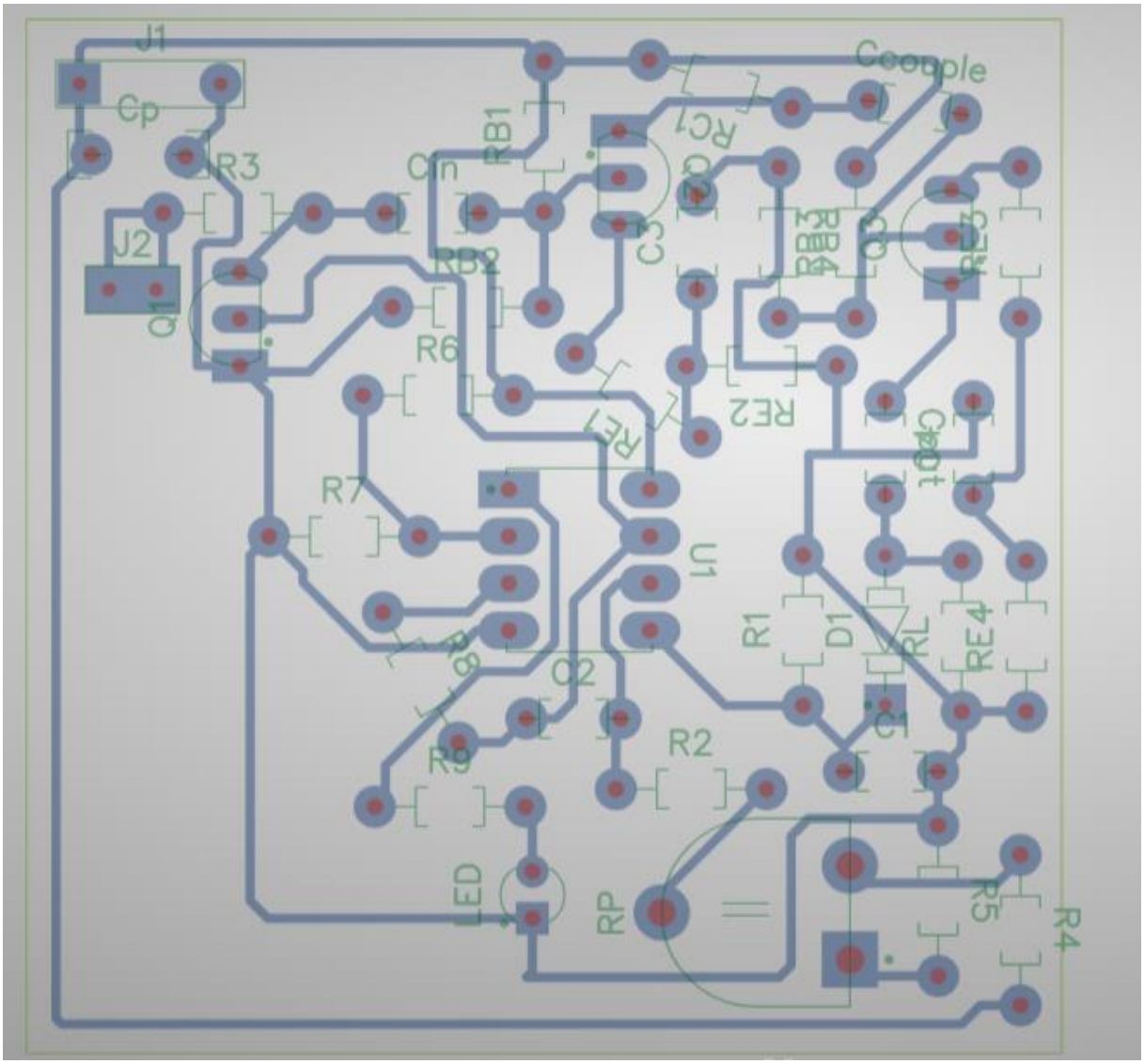

Figure 1 Layers photo

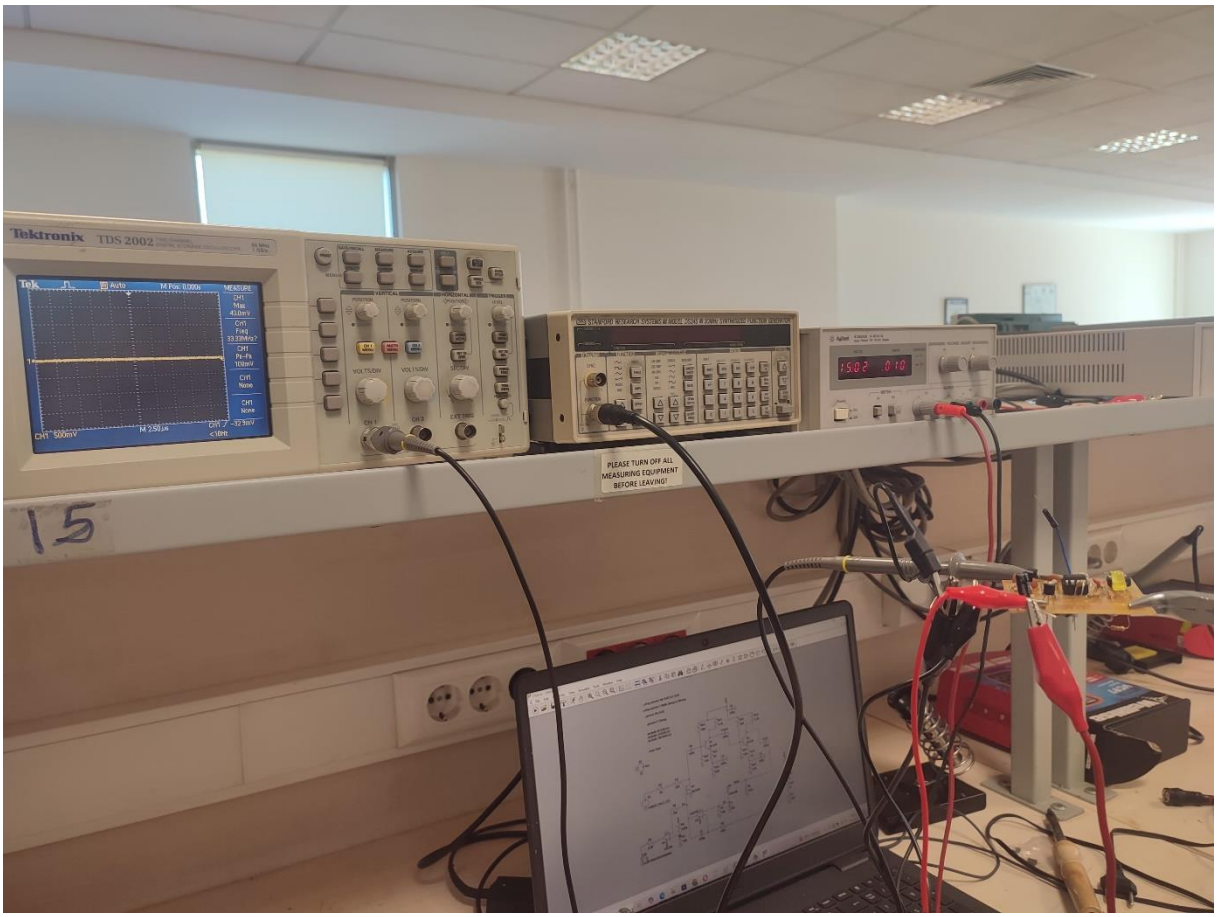

Figure 2 When input is less than 10mV

When inpıt is less than 10 mV LED is OFF.

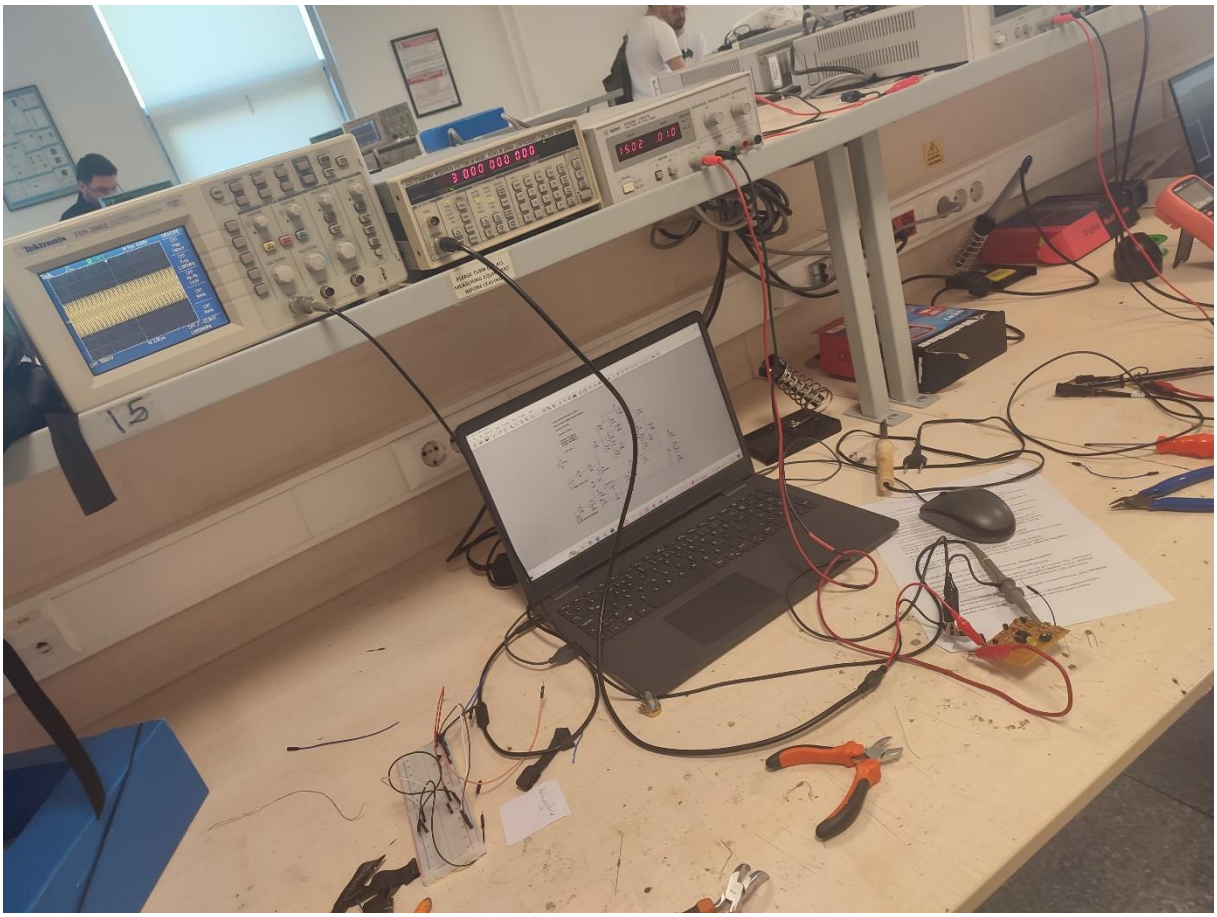

Figure 3 at 3Mhz

At 3 MHz LED is OFF.

Figure 4 200kHz and Vin=0.01

At 200KHz and Vin=0.01V so Vout=0.9V.

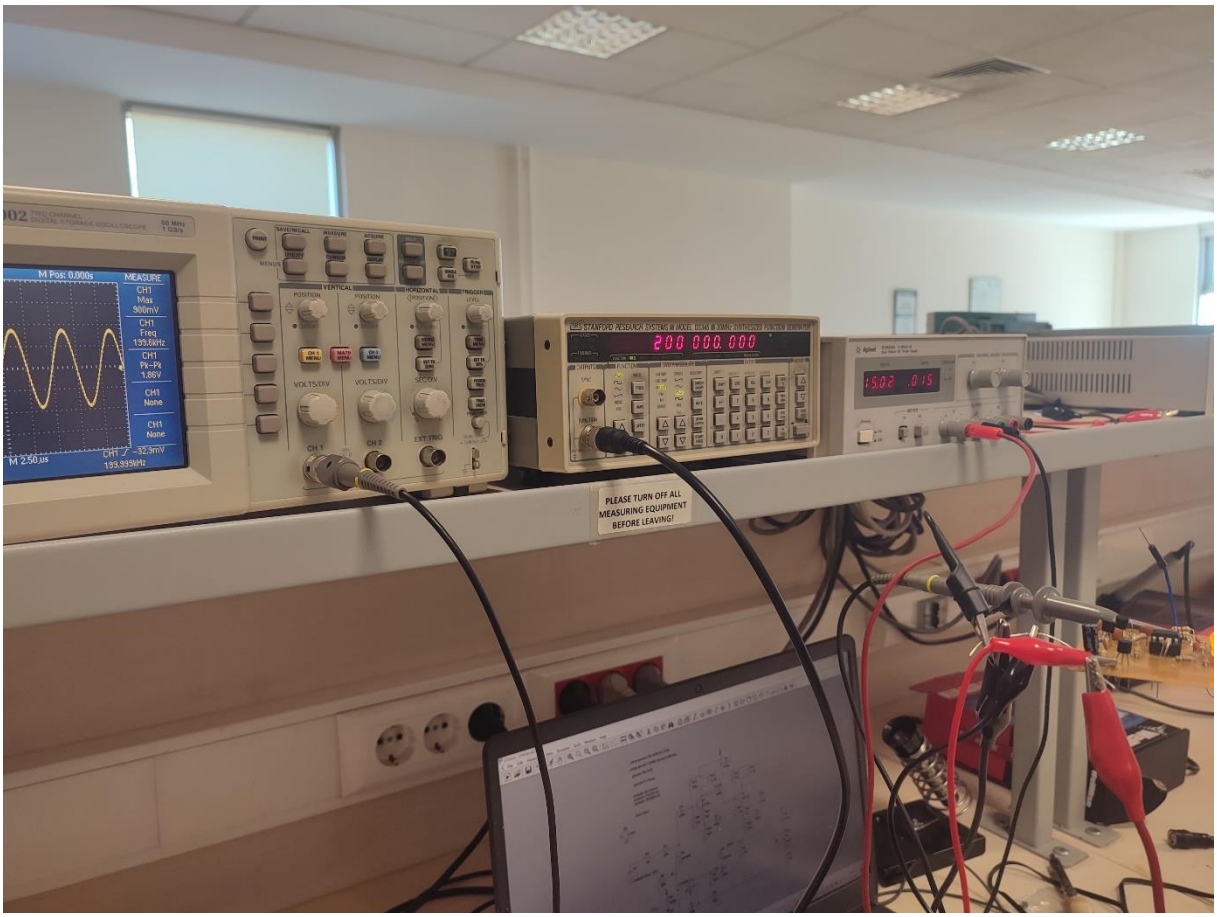

Figure 5 200kHz and Vin=0.1

At 200KHz and Vin=0.1V so Vout=0.9V.

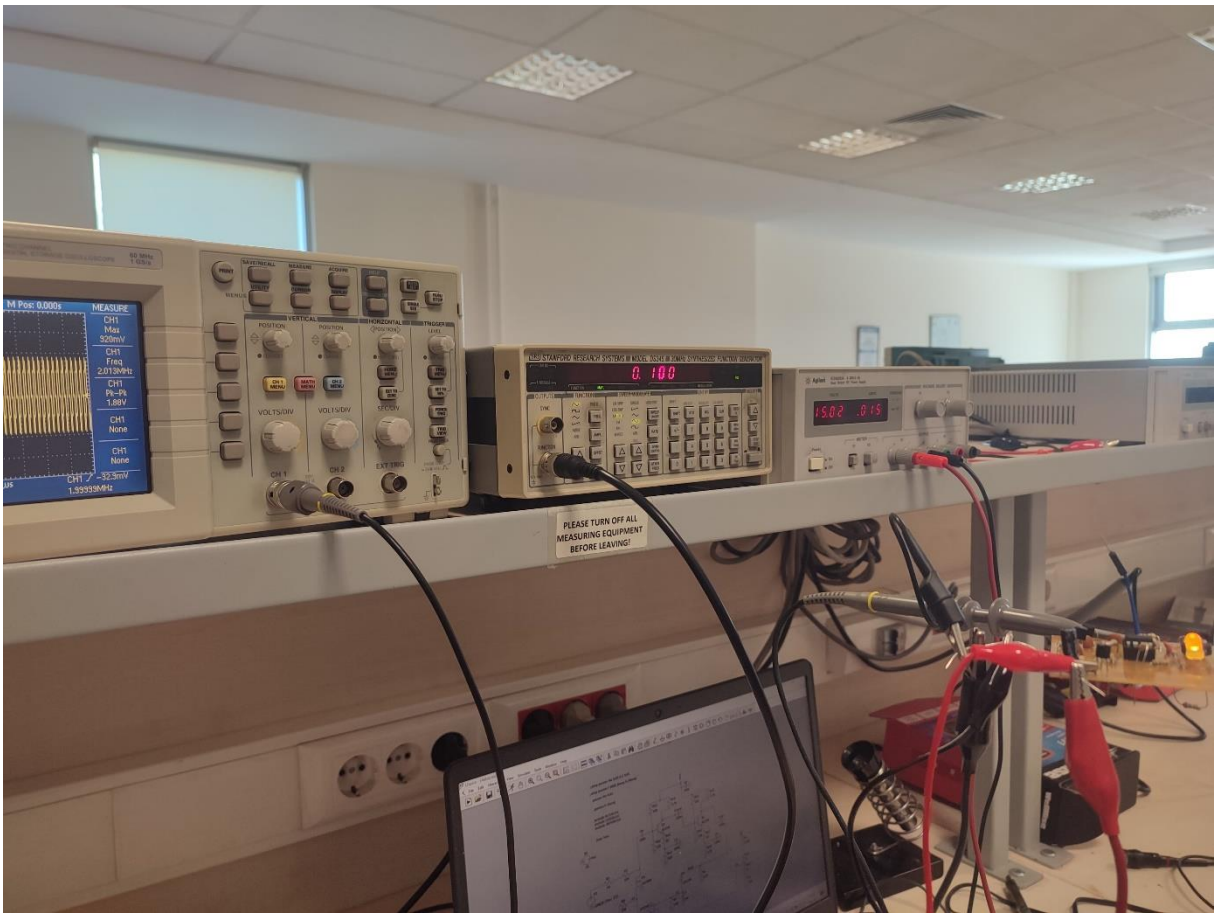

Figure 6 2MHz and Vin=0.1

At 2MHz and Vin=0.1V so Vout=0.92V.

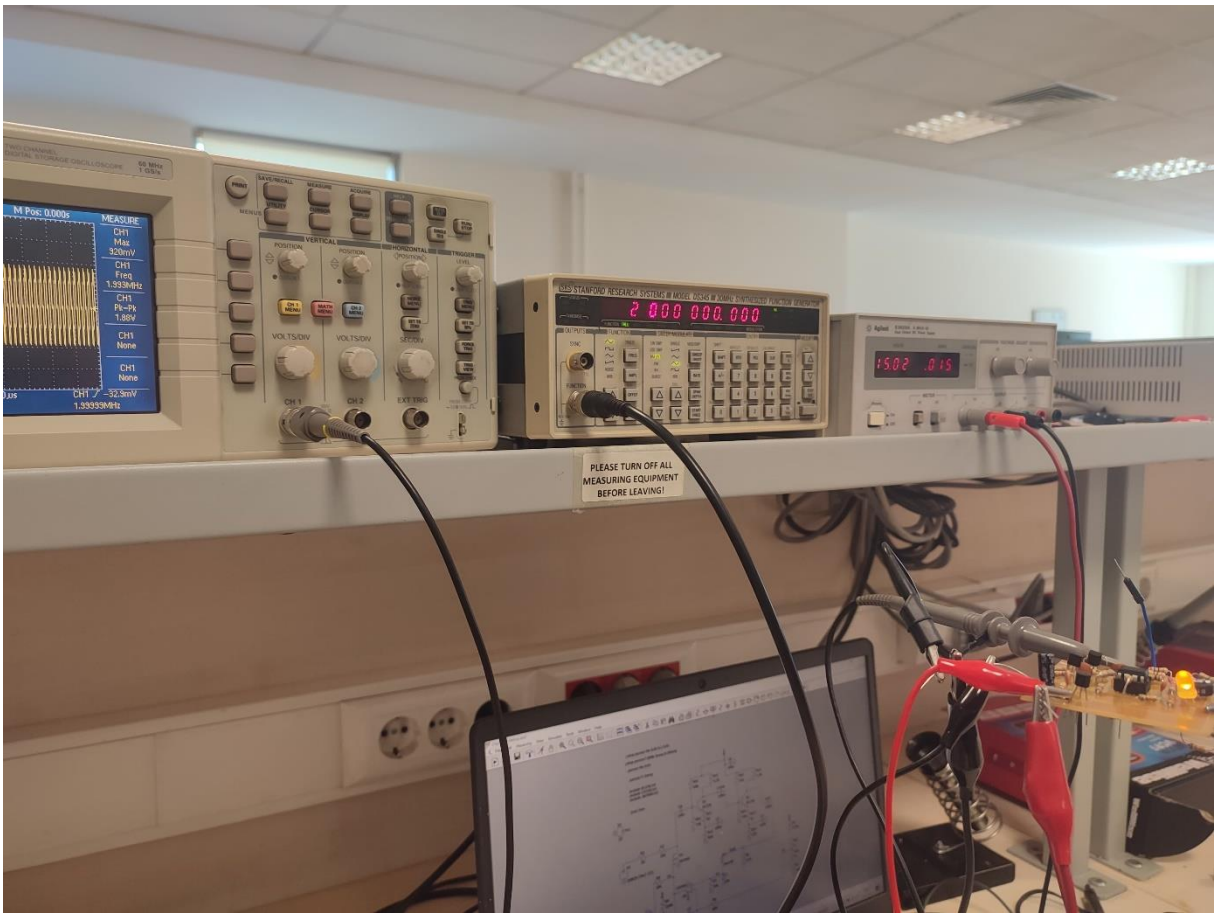

Figure 7 2MHz and Vin=0.01

At 2MHz and Vin=0.01V, so Vout=0.92V.

In the end, specification 1 is satisfied. Also, in every photo, Ic=15mA son the second specification is also satisfied. Finally, in Figure 2, you can see the LED is OFF when the input is less than 10 mV.

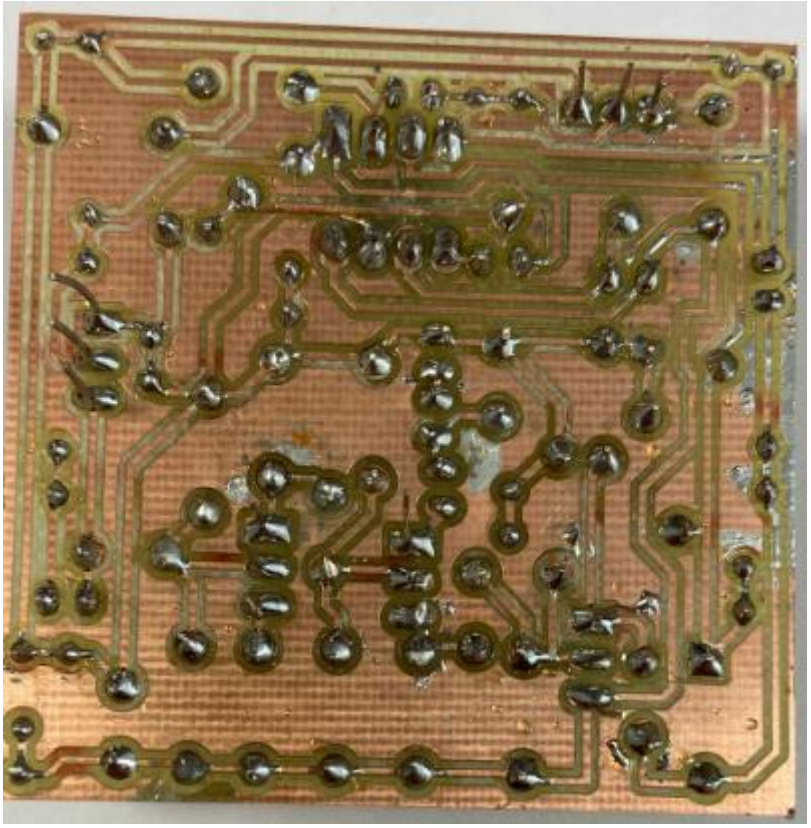

Figure 8 back side of pcb

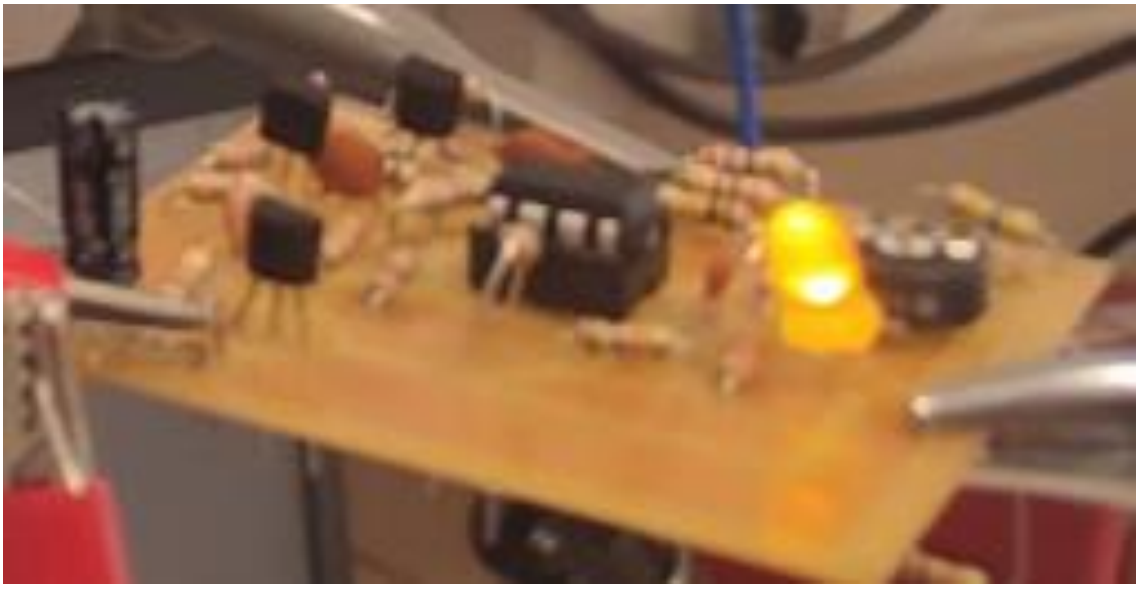

Figure 9 Pcb from front

Figure 10: other angle

In the PCB, I didn’t use any jumper wires. I have just used extra resistors to arrange some resistor values.